Component	Description/function	Significance
Airframe	Constructed using lightweight composite materials such as carbon fiber or fiberglass; optimized for aerodynamic efficiency.	Provided durability, increased endurance, and reduced aerodynamic drag, enhancing flight stability and performance.
Propulsion system	Included piston engines, small turbines, or electric motors, depending on mission type and payload requirements.	Enhanced power-to-weight ratio, improved speed, and extended operational endurance for diverse missions.
Flight controller (FCU)	The onboard microprocessor that processes sensor input and translates pilot or automated commands into control actions.	Served as the UAV's "brain," enabling semi-autonomous and stable flight operations through real-time data processing.
Global positioning system (GPS) module	Integrated satellite navigation receiver enabling waypoint navigation, autonomous flight paths, and return-to-home functions.	Allowed precise, repeatable, and autonomous flight, significantly improving navigational accuracy and mission reliability.
Inertial measurement unit (IMU)	Contained gyroscopes and accelerometers to measure motion, orientation, and angular velocity.	Enhanced flight stability, reduced operator input, and enabled smooth flight under variable environmental conditions.
Communication and telemetry system	Established a two-way data link between the UAV and ground control station (GCS)	Enabled real-time transmission of telemetry, video, and